Question 3 — Dynamic Task Scheduling & Timer Library Coordination

Timer Configuration

The Timer.h library is used to control the different timed tasks in the system.

- LED 1 toggles every 250 ms.
- LED 2 produces a 150 ms pulse every 3 seconds.
- A system counter increases every 1000 ms.
- The LCD is refreshed every 250 ms.

Timing Calculations

LED 1 changes state every 250 ms, giving it a complete ON/OFF cycle of 500 ms.

LED 2 is activated once every 3000 ms and remains active for 150 ms. After the pulse ends, the LCD displays "PULSE: EXPIRED" for another 450 ms.

This means the pulse-related LCD message is handled for:

- 150 ms while the pulse is active.
- 450 ms after the pulse ends.

The extra timing is controlled using millis() without using delay() or creating another timer.

System Behavior

The timer library allows several tasks to run at the same time without stopping the program.

The LCD displays:

- Row 1: The current system tick count.
- Row 2: "PULSE: ACTIVE" while LED 2 is ON, then "PULSE: EXPIRED" for 450 ms after the pulse ends.

All timing tasks continue to operate independently and without blocking the main program.